

LIKITH SAI DHEEKONDA

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QUALIFICATIONS SUMMARY

Experienced engineer specializing in the design and programming of embedded systems, with expertise in Baremetal, HAL, RTOS, and Linux drivers. Skilled in seamlessly integrating coding, hardware, and system design to create efficient and robust solutions. Highly adaptable and adept at tackling multidisciplinary challenges with precision and innovation.

EMPLOYMENT EXPERIENCE

Hyderabad Tulaman Pvt Ltd

Senior Manager - R&D Electronics

Design & Development of Industrial grade Weighing scales, Weigh bridge, Weigh in Motion systems.

January 2024 - Current

Random Technical Systems

Managing Partner - Civil and Defence industry Subcontractor

Prototype development, System Deployment, Feasibility study

August 2021 - August 2023

Capgemini

Senior Software Engineer

ABAP Consultant

June 2016 - December 2017

EDUCATION

International Institute of Information Technology Hyderabad

Research Scholar, Electronics & Communication Engineering

Research in Speech technologies and Signal Processing

Developed embedded speech synthesizer and speech enhancer

December 2017 - December 2021

VNR Vignana Jyothi Institute of Engineering and Technology

B.Tech. in Electronics & Communication Engineering

Engineer in Research and Consultancy Cell

Overall GPA: 7.77

2012-2016

ACADEMIC RECOGNITION

Acknowledged contributor in IEEE Transactions on Instrumentation and Measurement Vol 75, 2026 journal for Vision-Enabled Smart Measurement Node for Hive Weight and Comb-State Monitoring.

PROJECT EXPERIENCE

Weapons locking and tracking system

Description: A battery powered embedded system module to track and secure high powered rifles in field, remotely.

Role: Electronics engineer, embedded engineer

Funded by ARDE - DRDO, Pune

Shot counter

Description: A battery powered embedded system module to count number of shots fired by a handheld weapons by shock wave signal processing.

Role: Electronics engineer, embedded engineer, Signal processing engineer

Funded by ARDE - DRDO, Pune

Digital Dendrometer

Description: An IoT based very low powered high precise embedded system to monitor the change of tree stem radius(max change over 24hrs is 100um) to track a plantation health

Role: Electronics engineer, Embedded engineer, Signal processing engineer, Instrumentation engineer, Cloud architect

Funded by private engineering firm.

Millibots

Description: Development of tiny light weight robot(under 4gms including battery).

Role: Electronics engineer, embedded engineer, RF engineer

Funded by DYSL - DRDO, Pune.

Speech data collection project

Description: A platform to collect, process speech data through user contributed call recordings.

Role: Android application devloper, Cloud Architect, Signal processing engineer

Managed by Prof Raj Reddy, Carnegie Mellon University.

TECHNICAL STRENGTHS

Programming Languages	C/C++, Assembly, Java, MATLAB, Python, HTML, CSS, Javascript
Computer Technologies	Android app development, Machine Learning & Neural networks, Speech processing, Image processing, Cloud engineering(GCP, Azure)
Embedded Technologies	Baremetal, FreeRTOS, Embedded linux, Embedded ML, Embedded Signal processing, FPGA synthesis, Reverse engineering(GHIDRA)
Embedded Platforms	ARM programming(STM, TI, NXP, Cypress, Espressif, Nordic, RPI PICO, MICROCHIP, Mindmotion, Puya Semi, Nuvoton Arduino), RISC-V, SOM & SBC
Embedded Peripherals	Device Driver, PIO, UART, USART, SPI, I2C/TWI, CAN, ADC, DAC, PWM, ICR, DMA, USB, IR (RC5, NEC), MODBUS
Electronics design	Analog system design, Digital system design, Power supply design, VLP(Very low power) circuit design, RF & antenna design
Wireless Technologies	Bluetooth low energy 5.2, WIFI (802.11 b/g/n), lora, GPS, RFID (UHF), QI(wireless charging)
CAD platforms	Kicad(PCB design), Solidworks, Freecad(Mechanical design), Autocad
FAB skills	Hand soldering(starting from 0201 size components), PCB assembly, 3d printing, laser cutting Turnkey prototyping
Other skills	Electrical engineering, Control sytems engineering, Robotics, FEA/FEM, Latex scripting